

# MARK LORENZ YAMANAKA

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## SUMMARY

PRC-licensed Professional Aeronautical Engineer (2024) pursuing a career in aircraft structures and stress engineering. Background in airframe maintenance and structural analysis, with hand calculations as the baseline for FEA and any result that looks too good checked before it is trusted. Looking to join a stress team, learn from experienced engineers, and grow into detailed substantiation and certification work.

## SKILLS

**Stress & Structural Analysis:** Hand calculations, stress linearization, membrane-plus-bending classification, V-n envelope, limit and ultimate load, mesh convergence, sensitivity studies

**FEA & CFD:** ANSYS Mechanical, ANSYS Fluent, one-way FSI, CalculiX, PrePoMax, OpenFOAM, ParaView, XFOIL

**CAD & Drawings:** CATIA V6, SolidWorks, ANSYS SpaceClaim, GD&T, tolerance stack-up, BOM

**Standards:** FAR Part 23 / 27 / 29, ASME Y14.5-2018, ASME BPVC VIII-1 Appendix 2, PCAR Part 6, FAR Part 145

**Documentation:** Stress reports, substantiation packages, task cards, work orders, non-routine cards, certificate of release to service

**Programming & Tools:** Python, MATLAB/Simulink, C++, Git/GitHub, Microsoft Office, Power BI, Asana, OriginLab

**Languages:** English (Native), Cebuano (Native), Filipino (Professional), Japanese (Conversational)

**Certifications:** PRC Professional Aeronautical Engineer License, Philippines, December 2024

## EXPERIENCE

### Aircraft Maintenance Engineer Intern | Dornier Technology Inc.

May 2023 – Jul 2023

- Supported 10+ base and line maintenance checks at a third-party MRO on customer fleets (Cebu Pacific, Royal Air) covering A320, ATR 72-500, B737-200 and BAe 146, under PCAR Part 6 AMO approval, the Philippine equivalent of FAR Part 145.
- Performed 15+ pre- and post-check inspections during line maintenance, then supported an A320 engine overhaul in base maintenance.
- Installed torque-controlled fasteners on an A320 wing leading-edge de-icing boot under a licensed mechanic's sign-off; assisted access panel removal and refit throughout.
- Inspected in-service airframe structure for damage and corrosion; observed and assisted dye penetrant and borescope inspection.
- Handled the records chain for each task: task cards, work orders, non-routine cards raising unplanned findings, and certificate of release to service; documentation closed internal audit with zero non-conformities.
- Reported to licensed aircraft maintenance engineers and base maintenance supervisors.

### Ground Support Group Intern | Alpha Aviation Group Philippines

Feb 2023

- Completed flight simulation sessions on a Cessna 172 and an Airbus A320 full-flight simulator during a two-week rotation.
- Performed external inspection and cockpit preparation on trainer aircraft.
- Conducted a supervised engine run-up and taxi on the Cessna 172, working through the function checks and fault indications that accompany them.

### Part-Time Leader / Sub-Leader, Packaging and Logistics | ZOZOBASE

Feb 2018 – May 2026

- Led shift operations for a 15-member packaging and logistics team in a Japanese-language environment, briefing floor operators and reporting to supervisors.
- Proposed workflow changes adopted by shift supervisors, raising throughput 20% and cutting cycle time 12%.

## PROJECTS

### Uncooled SS316 Convergent-Divergent Nozzle | [github.com/dmmarkwork-cloud/ss316-cd-nozzle-feasibility](https://github.com/dmmarkwork-cloud/ss316-cd-nozzle-feasibility)

- Carried an uncooled SS316 C-D nozzle ( $P_c = 2$  MPa,  $T_c = 800$  K) from CFD through thermal-structural analysis into a full GD&T drawing set. One-way FSI mapped Fluent wall temperature and pressure onto the FEA mesh; linearized membrane-plus-bending stress placed the governing location at the chamber-to-converging junction (29.6 MPa) rather than the throat (17.2 MPa), giving FoS 7.12 across a three-level mesh convergence study.
- Passed CFD validation gates at peak wall  $y^+ 1.74$ , mass imbalance  $3.0e-5\%$ , discharge coefficient 0.977, and exit Mach 2.85 against 2.94 by hand calculation. Identified creep at a 527 °C wall, not static yield, as the real design driver.

- Sized an 8× M8 A286 bolted joint per ASME BPVC VIII-1 Appendix 2 and detailed the package in CATIA V6 with tolerance stacks closing at 1.68×, machining route, inspection plan and BOM. Published the v1.0 error (FoS 42.70, traced to a wrong assessment location and stress misclassification) alongside the corrected v2.0.

#### **AI 6061-T6 Propellant Tank: Static FEA Verification** | [github.com/dmmarkwork-cloud/gas-vessel-fea](https://github.com/dmmarkwork-cloud/gas-vessel-fea)

- Ran static linear FEA of a 3 mm-wall tank with hemispherical caps at 3.25 MPa (inner radius 147 mm, R/t = 49) in CalculiX via PrePoMax using C3D10 elements. Mid-cylinder von Mises 136.6 MPa against 137.91 MPa by thin-wall hand calculation, agreeing to 1.0%, with mesh converged at 0.62% on 264,007 nodes.
- Reported FoS 1.91 against a self-defined 2.0 target at the cylinder-to-hemisphere junction, a 3D geometric effect thin-wall theory does not capture, and recommended a 3.2 mm wall. Documented an April 2026 boundary-condition correction that moved peak stress from 204.1 MPa to 144.5 MPa by relocating constraints off the critical junction.

#### **MK-10 Light Aircraft: FAR Part 23 Design & Structural Sizing** | [github.com/dmmarkwork-cloud/mk10-aircraft-design](https://github.com/dmmarkwork-cloud/mk10-aircraft-design)

- Held one design baseline from mission definition through weight and CG, drag buildup and the V-n envelope into spar sizing for a two-seat light aircraft (1,649 lb gross weight, 126.5 ft<sup>2</sup> wing, aspect ratio 6.58, NACA 2415, 125 BHP). Validated 459 nmi range, 7.63 h endurance and 9,536 ft service ceiling.
- Sized front and rear wing spars from a 10-strip spanwise load distribution per Bruhn to a 4.4G utility limit and 7.4G ultimate, carrying every weight change through to the shear-moment diagrams, then resolved the design into control surfaces, landing-gear load cases and a SolidWorks three-view.

#### **PANO-VISTA: Fixed-Wing UAV Design & Systems Integration** | [github.com/dmmarkwork-cloud/pano-vista-uav](https://github.com/dmmarkwork-cloud/pano-vista-uav)

- Modelled a tractor-configuration fixed-wing UAV part by part in one SolidWorks master assembly, then built it with brushless propulsion (A2212/6T 2200KV, 40A ESC, LiPo), an eight-channel receiver and four micro servos across three control axes. Delivered exploded isometric and full orthographic drawings with BOM, integrating vendor CAD alongside custom parts. Five-person team project.

#### **Wooden Rotor Blade: Drawing Package** | [github.com/dmmarkwork-cloud/rotor-blade-helicopter-design](https://github.com/dmmarkwork-cloud/rotor-blade-helicopter-design)

- Drafted a Ø 71.10 in two-blade helicopter rotor in orthographic projection on tracing paper, then rebuilt it parametrically in SolidWorks as a complete blade and hub assembly with dimensioned three-view and isometric. Nonlinear twist falls from 78° inboard to 24° at the tip so angle of attack holds roughly constant as local velocity climbs with radius; solidity 0.095.

#### **High-Lift Device Demonstrator: Servo-Actuated Wing Section**

- Built a sectioned wing in cardboard skinned in aluminium foil on a presentation stand, with servos bedded between the ribs driving the ailerons and leading-edge slats from an Arduino Uno sketch written in C++, operated from a dedicated control interface on the base so the high-lift devices deploy on demand at eye level.

## **ORGANIZATION**

### **Chairperson / Officer-in-Charge** | **SEA-HAU Debate Committee**

**Sep 2022 – Nov 2022**

- Led a university-wide debate program with 25+ participants, coordinating judges and partner organizations from planning through execution.
- Developed standardized evaluation criteria and topic frameworks to keep judging fair and consistent across the competition.

### **Point-Person** | **SEA-HAU Debate Committee**

**Jul 2022 – Aug 2022**

- Acted as primary liaison between debate participants and the organizing committee, relaying feedback and enforcing committee policy.
- Guided participants on parliamentary debate procedure and coordinated between teams, adjudicators, and committee members during events.

## **EDUCATION**

### **Bachelor of Science in Aeronautical Engineering**

**Jul 2020 – May 2024**

Holy Angel University, Angeles City, Philippines

- Honors: Dean's List (GPA 3.6/4.0)
- Relevant Coursework: Aircraft Structures, Aircraft Systems, Aircraft / Helicopter / UAV Design, Aerodynamics, Powerplant, Reliability Engineering, Engineering Data Analysis